

1. Objective / Problem Statement

Develop a system that **accurately estimates traffic volume** in real-time or for future prediction using **machine learning techniques**. This can be used by:

- Traffic authorities for management
 - Navigation apps for routing
 - City planners for infrastructure development
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2. Key Features / Ideas

Area	Ideas
Data Collection	Use data from sensors, traffic cameras, GPS, Google Maps API, open datasets (e.g., Caltrans, INRIX, METR-LA, etc.)
Data Types	- Time-series sensor data - Video feeds (convert to count using CV models) - Weather info - Holiday/calendar data - Road construction data
Prediction Output	- Traffic Volume (vehicles/hour) - Congestion Level (Low/Med/High) - Expected Delay
ML Tasks	- Regression (volume) - Classification (congestion levels) - Time Series Forecasting
Real-time Component	Live dashboards using WebSockets or MQTT for real-time volume updates
Visualization	Heatmaps of traffic density, line charts, congestion alerts

3. Machine Learning Ideas

► Data Preprocessing

- Handle missing values
- Normalize features
- Encode categorical data (e.g., weather, holiday)
- Create lag features for time-series
- Detect and remove outliers

► Modeling Approaches

Approach	Techniques
Traditional ML	Linear Regression, Random Forest, XGBoost, SVR
Deep Learning	LSTM, GRU, 1D CNN (for time series), MLP
Hybrid	Combine weather + temporal + location + traffic camera data
Computer Vision (if video used)	YOLOv8 or EfficientDet to count vehicles per frame
Forecasting	ARIMA, Prophet, DeepAR, TCN

4. Tech Stack

Category	Tools
Frontend (Dashboard)	React.js / Streamlit / Dash
Backend	Flask / FastAPI / Node.js
ML Libraries	Scikit-learn, TensorFlow, PyTorch, XGBoost
Visualization	Plotly, Seaborn, Leaflet.js (for map overlays)
Data Pipeline	Pandas, NumPy, Apache Kafka (for real-time), Airflow
Cloud Deployment	AWS EC2 / Heroku / GCP with CI/CD
Database	PostgreSQL / MongoDB / InfluxDB for time series

5. Datasets to Explore

- METR-LA Dataset (Traffic in LA)
 - PeMS Traffic Dataset (Caltrans)
 - City of Chicago Traffic Data
 - Google Open Traffic Data
 - Kaggle datasets: e.g., **Metro Interstate Traffic Volume**, **Traffic4Cast**
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6. Evaluation Metrics

Task	Metrics
Regression	MAE, RMSE, R^2
Classification	Accuracy, F1-score, ROC-AUC
Time-Series	MAPE, RMSE, Horizon Error Analysis

7. Possible Add-ons

- **Anomaly Detection** for accidents
 - **Route Suggestion Engine** based on predicted volume
 - **Event-aware Prediction** (e.g., stadium events, strikes, etc.)
 - **Mobile App** integration or alert system
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8. Folder Structure Idea

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TrafficTelligence/

├── data/

```
|   └─ raw/ processed/
|   └─ notebooks/
|   └─ models/
|   └─ src/
|   └─ preprocessing/
|   └─ model_training/
|   └─ prediction_service/
|   └─ dashboard/
|   └─ react_frontend/
|   └─ reports/
|   └─ requirements.txt
└─ README.md
```

□ 9. Use Case Flow

1. **Data Ingestion** → Sensors/CSV/API
2. **Preprocessing & Feature Engineering**
3. **Model Training & Evaluation**
4. **Deployment as REST API**
5. **Dashboard Visualization + Alerts**